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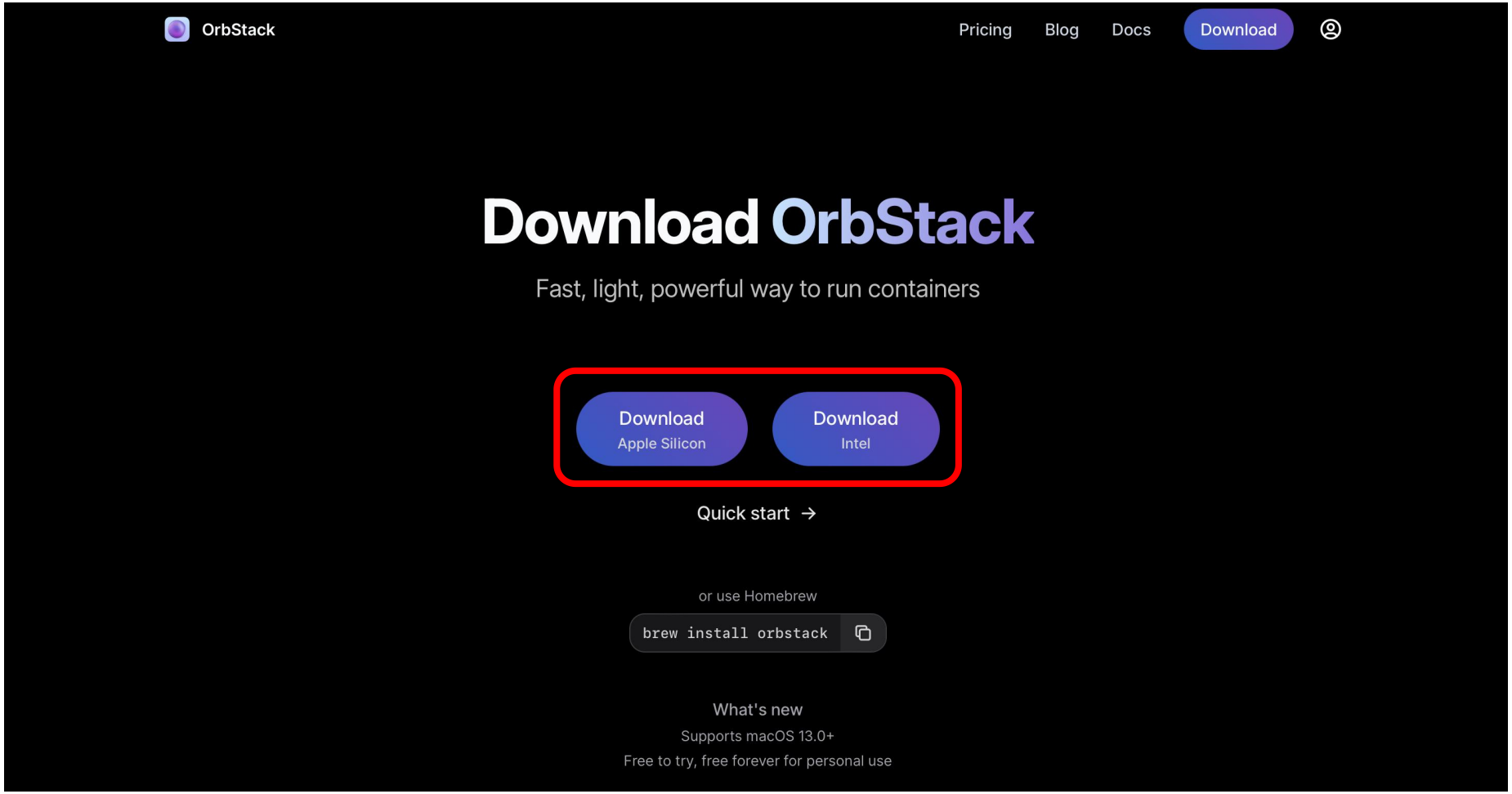
ECE2800J RC0 - Linux setup on your Mac

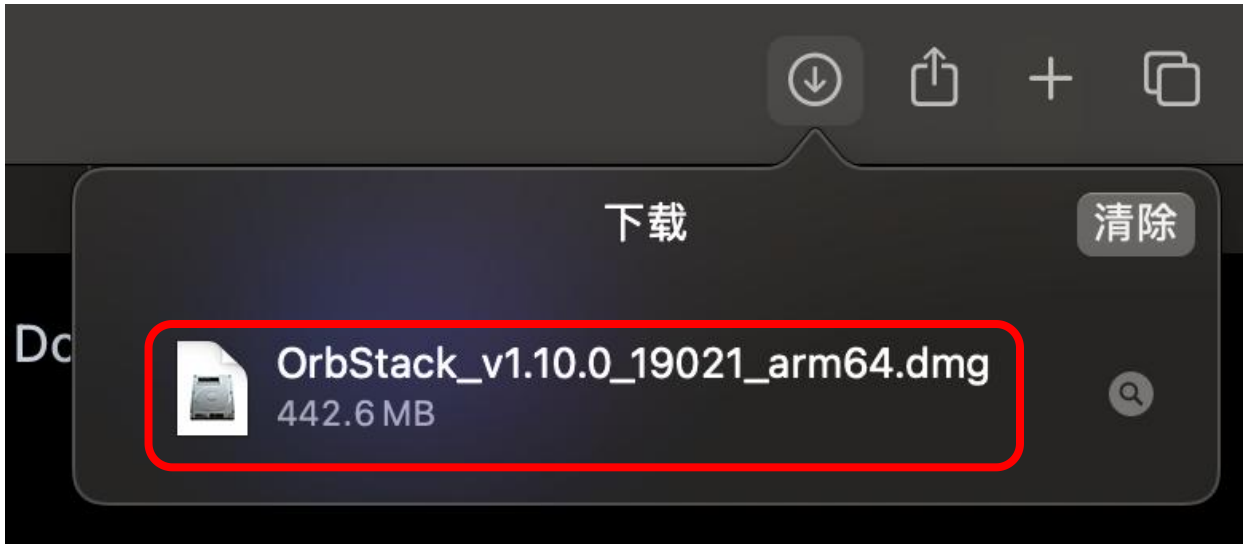
ji.sjtu.edu.cn

Check whether your Mac is with Apple Silicon or Intel.

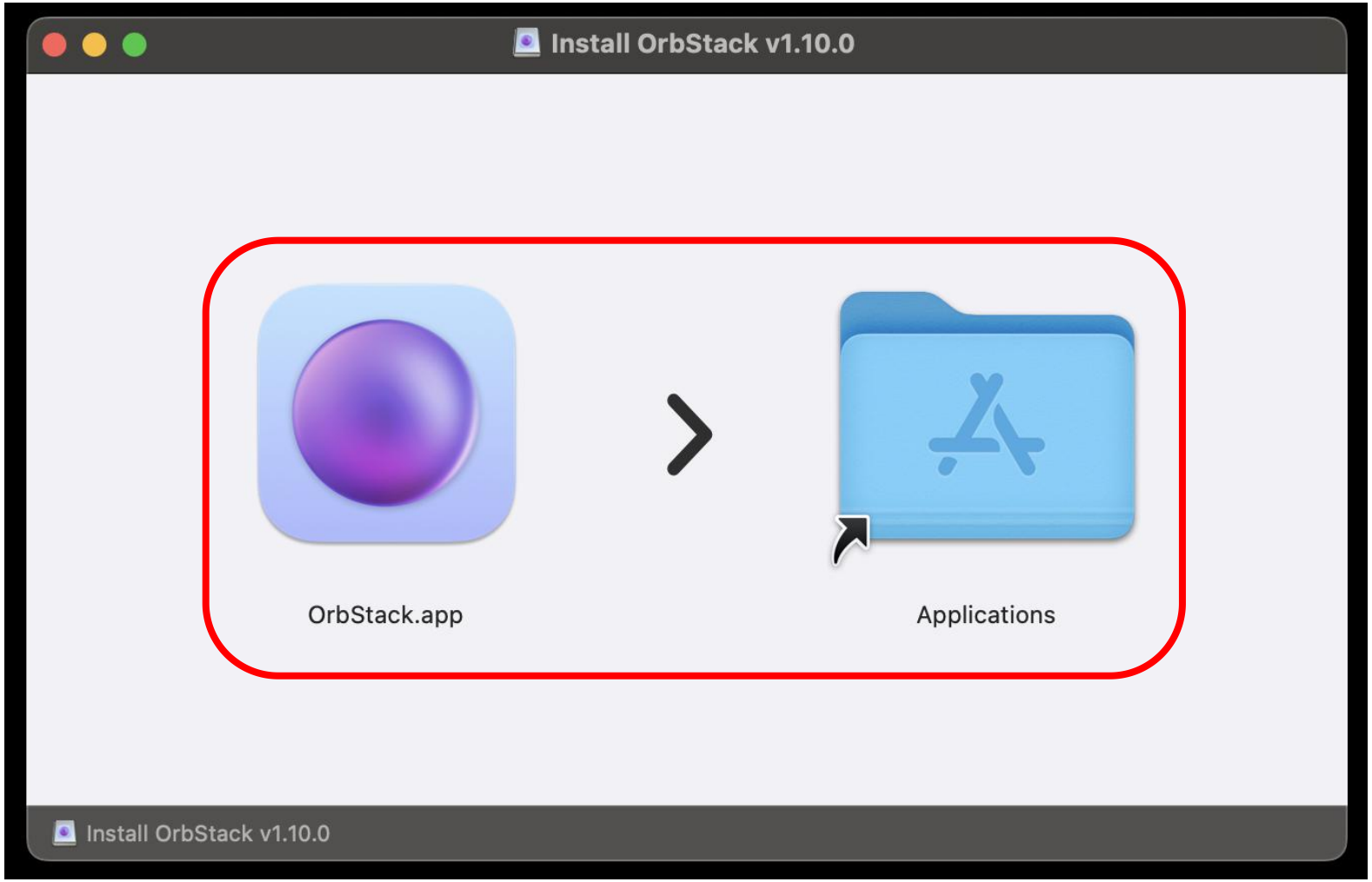
Go to https://orbstack.dev/download (Safari recommended), choose different download options according to your CPU type.



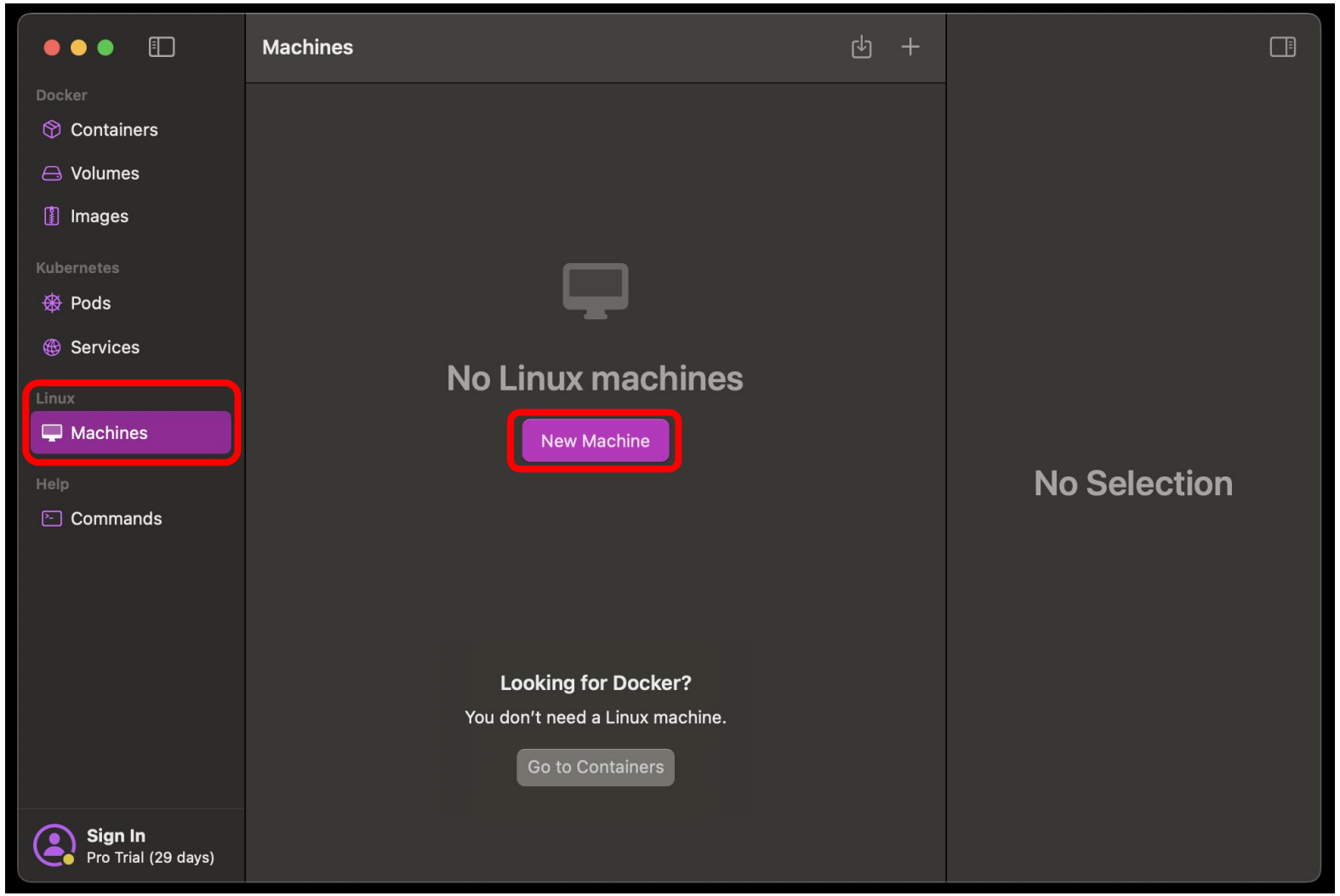
Take Apple Silicon as an example, double click the downloaded file.



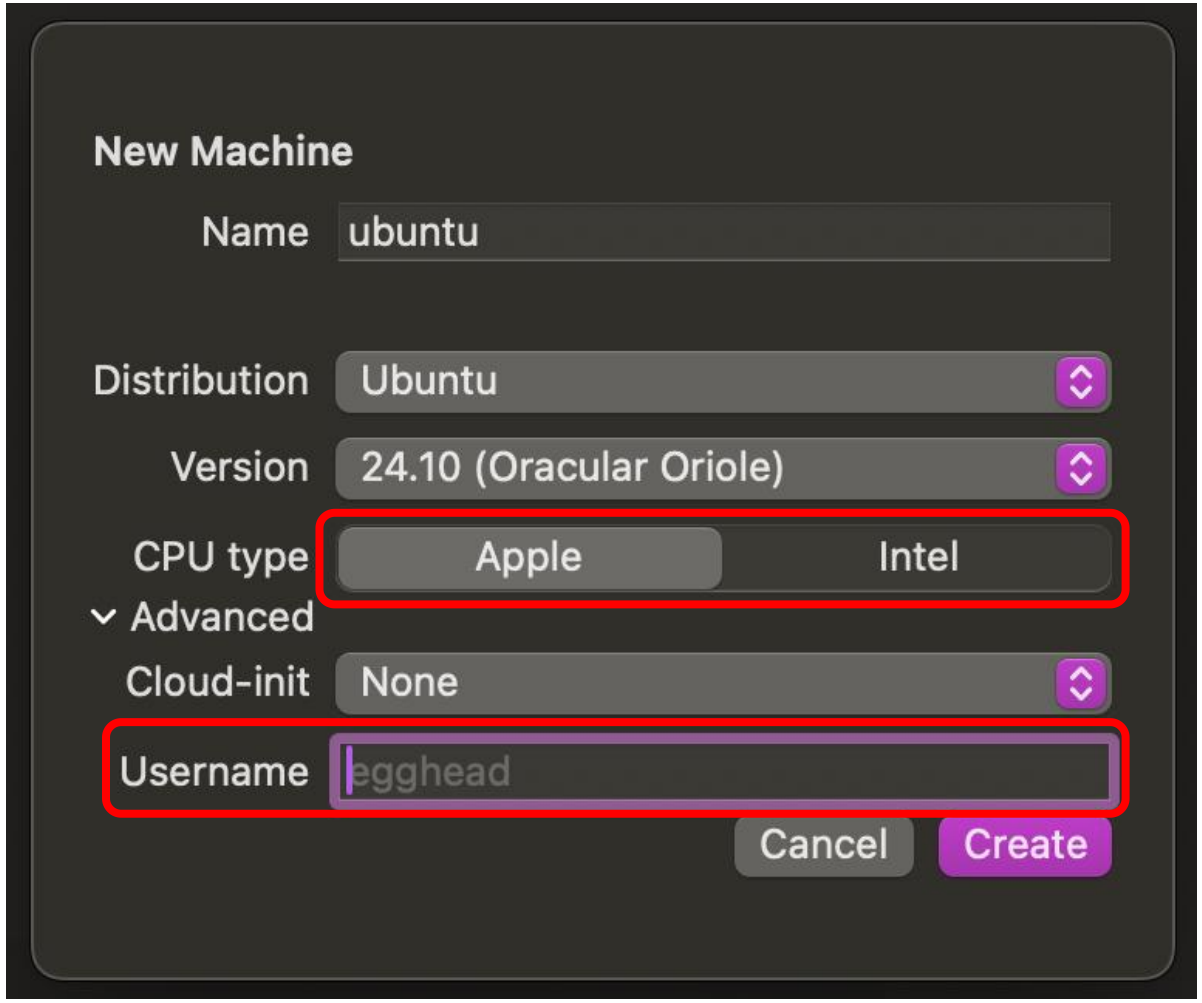
Move OrbStack into Applications.



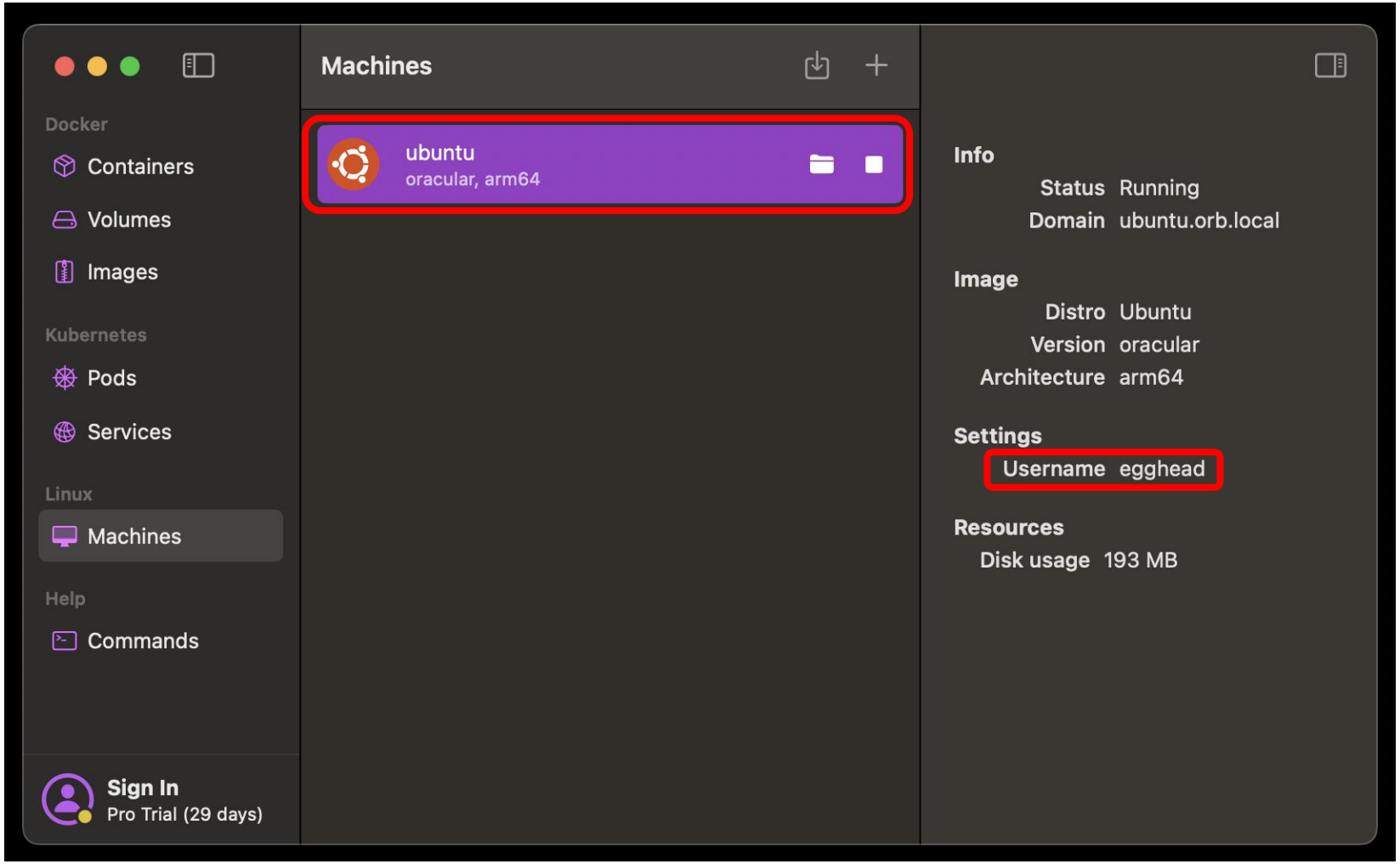
Now, you successfully download OrbStack.
Click OrbStack -> Machines -> New Machine



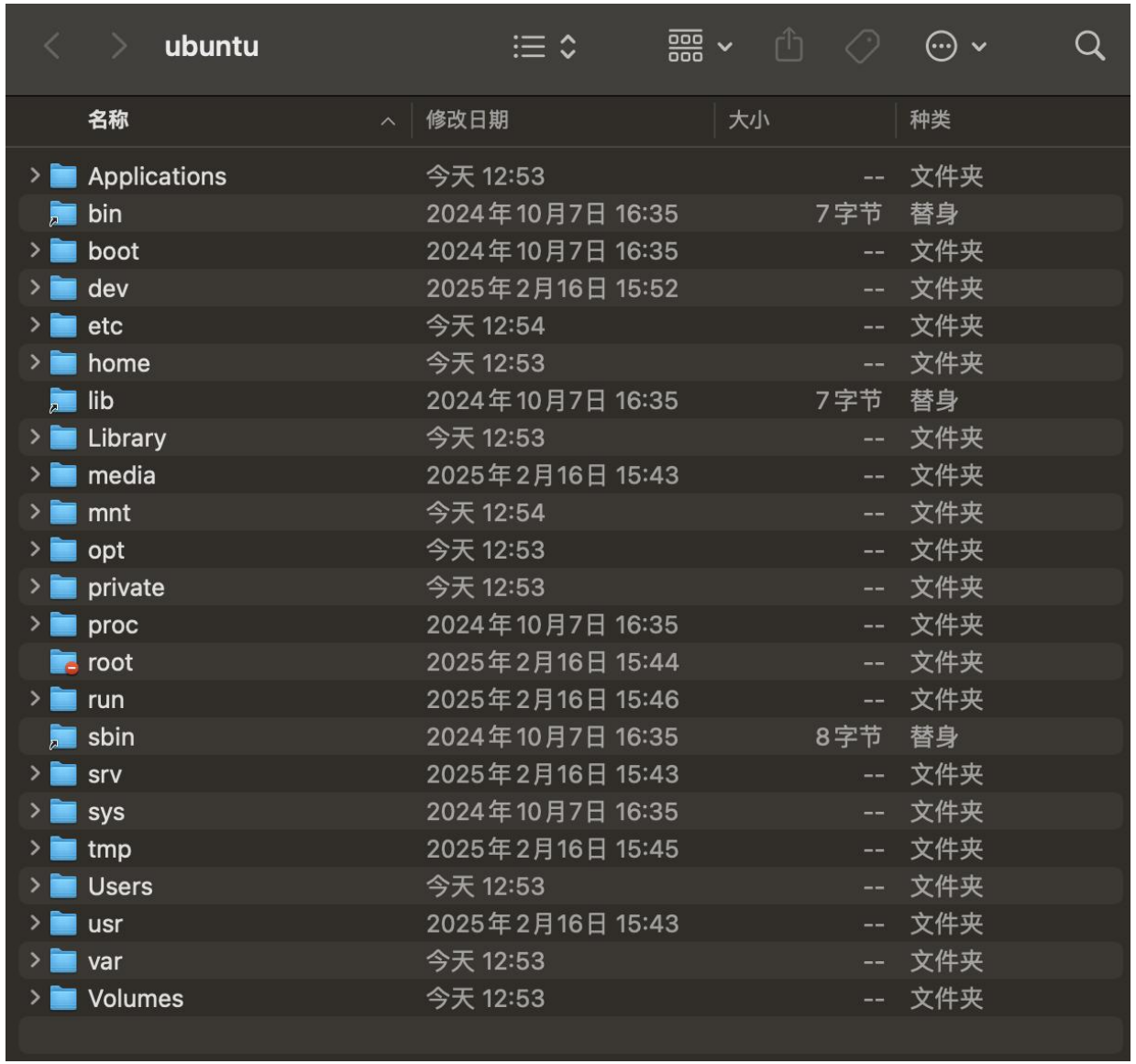
Then, you need to initialize your linux machine. Set as follows. You need to choose Apple or Intel according to your CPU type. Set the Username as you like. Then, click Create.



Wait for several seconds to allow your linux machine to get prepared. Then, you successfully initialize it.
The Username is set by you.



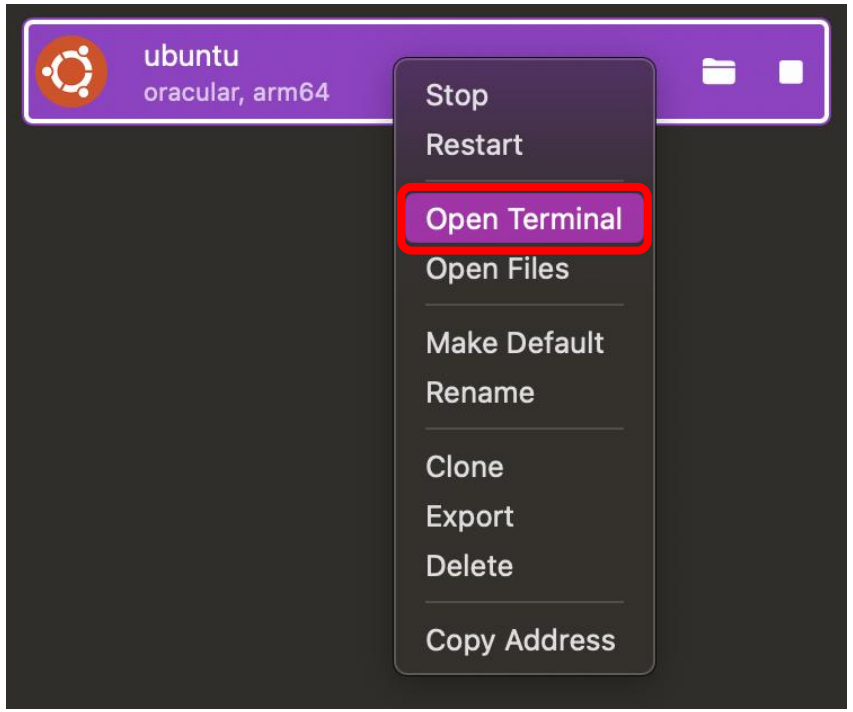
Click Open Files. All files related to your linux machine are shown.



Take a close look at home file. The user you just created is in here. Also, I recommend you to put all files related to ECE2800J under the newly created user (here is egghead) although you can use linux everywhere on your Mac.

Tips: on Mac, press Shift + Command + . to hide or show all files whose name starts with . .

Next, you need to know how to run linux on your Mac.
Right-click your machine and then Open Terminal.
Then, you open the home directory of your Mac by default, under linux environment.



Next, you need to install some tools which will be used in ECE2800J. These tools include gcc, g++, tree and valgrind. Vim has already been installed by default.

Run **sudo apt-get install gcc g++ tree valgrind** in the terminal.

To check whether you successfully installed them, run following commands one by one:

gcc --version

g++ --version

tree --version

valgrind --version

```
egghead@ubuntu:/Users/egghead$ gcc --version
gcc (Ubuntu 14.2.0-4ubuntu2) 14.2.0
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

egghead@ubuntu:/Users/egghead$ g++ --version
g++ (Ubuntu 14.2.0-4ubuntu2) 14.2.0
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

egghead@ubuntu:/Users/egghead$ tree --version
tree v2.1.1 © 1996 - 2023 by Steve Baker, Thomas Moore, Francesc Rocher, Flo
rian Sesser, Kyosuke Tokoro
egghead@ubuntu:/Users/egghead$ valgrind --version
valgrind-3.23.0
```

Now, everything is done!

I recommend you to write all code related to this class under a single directory, which is the user directory just created. In this case, do not forget to first cd to that directory.

Tips: If you want to get the absolute location of a directory, single-click the directory and press Option + Command + C to copy its location.

```
egghead@ubuntu:/Users/egghead$ cd /Users/egghead/OrbStack/ubuntu/home/egghead
egghead@ubuntu:/Users/egghead/OrbStack/ubuntu/home/egghead$
```

Advanced topics

- Why Orbstack?
- Other useful tools

Why Orbstack?

- Free and Light:
  - low CPU and disk usage when running
  - share files and dictories with MacOs
- Specially designed for Mac
  - work well on Mac with M1 and M2

Why OrbStack?

- ⚡ **Lightning Fast.** Starts in 2 seconds, optimized network and file system, fast x86 emulation.
- 💨 **Feather Light.** Low CPU and disk usage, battery-friendly, works with less memory, native Swift app.
- 🍰 **Effortlessly Simple.** Automatic domain names and migration, CLI & container/image/volume file access, VPN and SSH support.
- ⚙️ **Powerful.** Run Docker containers, Kubernetes, and Linux distros. Manage containers quickly from your menu bar. Explore volume and image files.

Other useful tools

https://iterm2.com

Other useful tools

https://developer.apple.com/xcode/

Xcode

Xcode enables you to develop, test, and distribute apps for all Apple platforms. Code and design your apps faster with predictive code completion, interactive previews, and live animations. Use Git staging to craft your next commit without leaving your code. Explore and diagnose your test results with redesigned test reports with video recording. And start deploying seamlessly to TestFlight and the App Store from Xcode Cloud. Creating amazing apps has never been easier.

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THANK YOU!

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